Week 5:

Grey wolf optimizer:

Application: Traffic signal timing optimisation

Formula:

$F = w_1 \cdot$ Average Waiting Time $+ w_2 \cdot$ Average Queue

Pseudo code:

1) Initialize parameters:
   - population size (number of search agents / wolves): N
   - maximum number of iterations: Max_iter
   - Search space bounds: [Min_Green_Time, Max_Green_Time] for each signal phase
   - Initialize wolf population x_i for $i = 1 \ldots N$

2) Evaluate initial objective function $F(x_i)$ for each wolf
   - Calculate Average waiting Time and Queue length for each timing vector x_i
   - Identify top three best solutions:
     * Alpha (x_{alpha}): Best search agent
     * Beta (x_{beta}): Second Best search agent
     * Delta (x_{delta}): Third best search agent.
     * Omega: last best

3) While iter < Max_iter do:
   For each search agent x_i do:
   - Update coefficient vectors a, A and C:
     * a decreases linearly from 2 to 0 over iterations
     * $A = 2 * a * r1 - a$
     * $C = 2 * r2$
   - Calculate vectors D_alpha, D_beta, D_delta for Alpha, Beta and Delta

- Calculate X_1, X_2, X_3 positions relative to Alpha, Beta, Delta, omega
- Update current search agent position: X_i = (X_1 + X_2 + X_3) / 3

End For

- Clamp positions X_i within valid green time bounds [min_green, max_green]
- Evaluate objective function F(X_i) for all updated search agents.
- Update Alpha, Beta, and Delta (omega) based on new objective values
- iter = iter + 1

End while

4) Return Alpha and its minimum cost

O/P:

```
-- Traffic Signal Timing Optimization using GWO --
Enter vehicle arrival rates (veh/sec) for 4 phases:
0.4 0.3 0.5 0.2
Enter saturation flow rates (veh/sec) for 4 phases:
0.9 0.8 0.9 0.85

-- Optimization Results --
Optimized Green Signal Timings for phases 1-4:
19.55 15.00 23.61 21.99
Total combined optimal Signal Cycle Time: 90.14 sec
Minimum Objective Cost Score: 14128.6245
```

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```python
import numpy as np


# Updated Objective Function with a total cycle length penalty
def traffic_objective_function(
    signal_timings, arrival_rates, saturation_flows
):
    """
    signal_timings: Array of green durations for each phase [G1, G2, G3, G4]
    arrival_rates: Vehicles arriving per second for each phase
    saturation_flows: Departure capacity per second of green time
    """
    total_waiting_time = 0.0
    total_queue_length = 0.0

    # Include yellow/all-red time (10 seconds total across phases)
    yellow_and_red = 10
    cycle_time = np.sum(signal_timings) + yellow_and_red

    # Define a minimum allowed total cycle time for a realistic urban intersection
    MIN_CYCLE_TIME = 90.0

    for i in range(len(signal_timings)):
        g = signal_timings[i]
        lambda_rate = arrival_rates[i]
        s_rate = saturation_flows[i]

        # Webster-inspired queue and delay model
        effective_red = cycle_time - g
        queue = max(0, lambda_rate * effective_red)

        # Average delay per vehicle in phase i
        delay = (cycle_time * (1 - (g / cycle_time)) ** 2) / (
            2 * max(0.001, 1 - (lambda_rate / max(0.001, s_rate * (g / cycle_time))))
        )
        delay = max(0, delay)

        total_waiting_time += delay * lambda_rate
        total_queue_length += queue

    # Calculate baseline traffic cost
    base_cost = 0.6 * (total_waiting_time / np.sum(arrival_rates)) + 0.4 * (
```

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```
    # Calculate baseline traffic cost
    base_cost = 0.6 * (total_waiting_time / np.sum(arrival_rates)) + 0.4 * (
        total_queue_length
    )

    # PENALTY: If the total cycle time is too short, add a heavy penalty
    if cycle_time < MIN_CYCLE_TIME:
        # Penalizes solutions proportionally to how far below the limit they are
        penalty = (MIN_CYCLE_TIME - cycle_time) * 1000
        return base_cost + penalty

    return base_cost


def gray_wolf_optimizer(
    obj_func,
    arrival_rates,
    saturation_flows,
    dim=4,
    search_agents_no=20,
    max_iter=50,
    lb=15,
    ub=60,
):
    # Initialize Alpha, Beta, Delta positions and scores
    alpha_pos = np.zeros(dim)
    alpha_score = float("inf")

    beta_pos = np.zeros(dim)
    beta_score = float("inf")

    delta_pos = np.zeros(dim)
    delta_score = float("inf")

    # Initialize population randomly within lower/upper bounds
    positions = np.random.uniform(lb, ub, (search_agents_no, dim))

    for iteration in range(max_iter):
        for i in range(search_agents_no):
            # Clip positions to hard boundaries
            positions[i] = np.clip(positions[i], lb, ub)
            fitness = obj_func(positions[i], arrival_rates, saturation_flows)

            # Update Alpha, Beta, Delta wolves
```

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```
            # Update Alpha, Beta, Delta wolves
            if fitness < alpha_score:
                delta_score = beta_score
                delta_pos = np.copy(beta_pos)
                beta_score = alpha_score
                beta_pos = np.copy(alpha_pos)
                alpha_score = fitness
                alpha_pos = np.copy(positions[i])
            elif fitness < beta_score:
                delta_score = beta_score
                delta_pos = np.copy(beta_pos)
                beta_score = fitness
                beta_pos = np.copy(positions[i])
            elif fitness < delta_score:
                delta_score = fitness
                delta_pos = np.copy(positions[i])

        a = 2.0 - iteration * (2.0 / max_iter)  # linearly decreases from 2 to 0

        for i in range(search_agents_no):
            for j in range(dim):
                r1, r2 = np.random.random(), np.random.random()
                A1 = 2.0 * a * r1 - a
                C1 = 2.0 * r2
                D_alpha = abs(C1 * alpha_pos[j] - positions[i, j])
                X1 = alpha_pos[j] - A1 * D_alpha

                r1, r2 = np.random.random(), np.random.random()
                A2 = 2.0 * a * r1 - a
                C2 = 2.0 * r2
                D_beta = abs(C2 * beta_pos[j] - positions[i, j])
                X2 = beta_pos[j] - A2 * D_beta

                r1, r2 = np.random.random(), np.random.random()
                A3 = 2.0 * a * r1 - a
                C3 = 2.0 * r2
                D_delta = abs(C3 * delta_pos[j] - positions[i, j])
                X3 = delta_pos[j] - A3 * D_delta

                positions[i, j] = (X1 + X2 + X3) / 3.0

    return alpha_pos, alpha_score
```

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```
    return alpha_pos, alpha_score


if __name__ == "__main__":
    print("=== Traffic Signal Timing Optimization using GWO ===")

    # Inputs matched to your configuration setup
    user_arrivals = input(
        "Enter vehicle arrival rates (veh/sec) for 4 phases separated by space [default: 0.4 0.3 0.5 0.2]: "
    )
    if user_arrivals.strip():
        arrival_rates = np.array([float(x) for x in user_arrivals.split()])
    else:
        arrival_rates = np.array([0.4, 0.3, 0.5, 0.2])

    user_sats = input(
        "Enter saturation flow rates (veh/sec) for 4 phases separated by space [default: 0.9 0.8 0.9 0.85]: "
    )
    if user_sats.strip():
        saturation_flows = np.array([float(x) for x in user_sats.split()])
    else:
        saturation_flows = np.array([0.9, 0.8, 0.9, 0.85])

    # Raised baseline bounds slightly and increased agent search scale for realistic coverage
    best_timings, best_cost = gray_wolf_optimizer(
        obj_func=traffic_objective_function,
        arrival_rates=arrival_rates,
        saturation_flows=saturation_flows,
        dim=4,
        search_agents_no=20,
        max_iter=50,
        lb=15,  # Shifted minimum individual floor to 15s
        ub=50,
    )

    print("\n--- Optimization Results ---")
    print(
        f"Optimized Green Signal Timings (seconds) for Phases 1-4: {np.round(best_timings, 2)}"
    )
    print(f"Total Combined Optimal Signal Cycle Time: {round(np.sum(best_timings) + 10, 2)} seconds")
    print(f"Minimum Objective Cost Score: {round(best_cost, 4)}")
```

```
=== Traffic Signal Timing Optimization using GWO ===
```

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```
        lb=15,  # Shifted minimum individual floor to 15s
        ub=50,
    )

    print("\n--- Optimization Results ---")
    print(
        f"Optimized Green Signal Timings (seconds) for Phases 1-4: {np.round(best_timings, 2)}"
    )
    print(f"Total Combined Optimal Signal Cycle Time: {round(np.sum(best_timings) + 10, 2)} seconds")
    print(f"Minimum Objective Cost Score: {round(best_cost, 4)}")
```

```
=== Traffic Signal Timing Optimization using GWO ===
Enter vehicle arrival rates (veh/sec) for 4 phases separated by space [default: 0.4 0.3 0.5 0.2]: 0.4 0.3 0.5 0.2
Enter saturation flow rates (veh/sec) for 4 phases separated by space [default: 0.9 0.8 0.9 0.85]: 0.9 0.8 0.9 0.85

--- Optimization Results ---
Optimized Green Signal Timings (seconds) for Phases 1-4: [19.55 15.   23.61 21.99]
Total Combined Optimal Signal Cycle Time: 90.14 seconds
Minimum Objective Cost Score: 14128.6245
```

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